

Singularity-free Collision Avoidance for UAVs in a 3D Obstacle-filled Environment

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Abstract—This research article introduces a new method of guiding Unmanned Aerial Vehicles (UAVs) through a static obstacle-filled environment in three-dimensional (3D) space. Traditional artificial potential field based approaches are extensively used for this problem but those methods have inherent deficiency of suffering from singularity which arises in the process of combining the main field with the obstacle field. To overcome this limitation, in this study a different approach is adopted, which involves modulating the main field in the presence of obstacles without adding it directly with the obstacle field. Additionally, nonholonomic constraints of the fixed-wing UAVs are also taken into account while navigating through obstacles. The streamlines of the proposed vector field give feasible near-optimal paths for UAVs, which can be easily tracked by any convergence guaranteed steering algorithm. Finally, the numerical simulations are presented to show the efficacy of the proposed approach.

I. INTRODUCTION

Major advancements in sensor technology over the last few decades, safety to human life and cost effectiveness have paved the way for autonomous systems to be used in a variety of fields [1] e.g. military and civilian sectors [2], [3], and [4]. In most of these applications, path planning problems have to be solved to generate a feasible path for UAVs to follow a set of waypoints, considering various constraints, such as, travel time, total distance to be travelled, continuity of the path profile and maintaining safe distance from obstacles. Navigation of UAVs with collision avoidance is prerequisite for the success of any such missions in the real-world applications and hence necessitates an extensive area of research. Due to unknown dynamics of intruders and the limited computational power of onboard processors computation of a safe trajectory in real-time for UAVs in an obstacle-filled environment is still a challenging problem.

Some of the well-known approaches of obstacle avoidance are grid search based algorithms ([5], [6]), road map based algorithms ([7], [8]), optimal control algorithms ([9], [10]), Artificial Potential Field (APF) algorithms ([11], [12]) *etc.* Except artificial potential field remaining all the approaches are computationally complex and require smoothing techniques after planning the path to make it implementable.

Khatib [11] proposed an artificial potential field approach for collision avoidance of robot manipulators where the goal

point is modeled as an attractive force and the obstacle is modeled as a repulsive force. Due to the inherent naturality and simplicity, APF based approaches have more potential in online trajectory planning. But the main drawback of the APF approach is that singularity of field occurs when an obstacle is exactly on the line between the vehicle and the goal point. Kim *et al.* [12] proposed a harmonic potential function which is derived as a solution of the Laplace equation with Dirichlet boundary conditions to avoid singularities in all the points in the field except at the goal and the source. Munasinghe *et al.* [13] presented an intelligent technique by modelling an obstacle as a velocity dipole to deal with the presence of moving obstacles effectively. Inspired by APF approaches, Li *et al.* [14] and Connolly *et al.* [15] proposed guidance algorithms using the concept of fluid mechanics. Poisson's equations are used in [14] to develop an ideal fluid to deal with the local minima or singularities. Here, streamlines are serving as obstacle free paths. Waydo *et al.* [16] developed streamline functions satisfying the Laplace equation for obstacle avoidance problems. Similar work was presented in [17] for 3D obstacle avoidance problems. Main advantages of streamline based approaches are that the resultant paths are smooth and practically realizable. Mainly these studies focus on modifying the actual vector field by combining it with the obstacle vector fields to realize both path-following and obstacle avoidance. In these cases, there is a possibility of cancellation of the main vector field by the obstacle field leading to singularities in those points [18]. At these singular points, the vehicles do not get any feasible speed and heading commands, which may cause failure of the mission.

Another approach similar to the APF method is the Artificial Velocity Vector Field (AVVF) method. In the APF approach, a negative gradient is used to locate local minima in the map, which is not required in AVVF methods. Wilhelm *et al.* [18] proposed a gradient vector field as a sink flow, where obstacles are modeled as circular fields. They employed parametric tuning using lookup tables to generate optimal paths that are smooth and suitable for curvature-constrained vehicles. Zhu *et al.* [19] investigated different scaling functions for modeling obstacles in an artificial vector field. Marchidan *et al.* [20] proposed collision avoidance vector fields using velocity modulation functions to generate constant velocity fields. Although singularities were addressed there but the minimum turn rate constraints of nonholonomic vehicles were not taken into account. Furthermore, all of these works are limited to 2D applications.

The novelty of this study is that (i) the singularity problem

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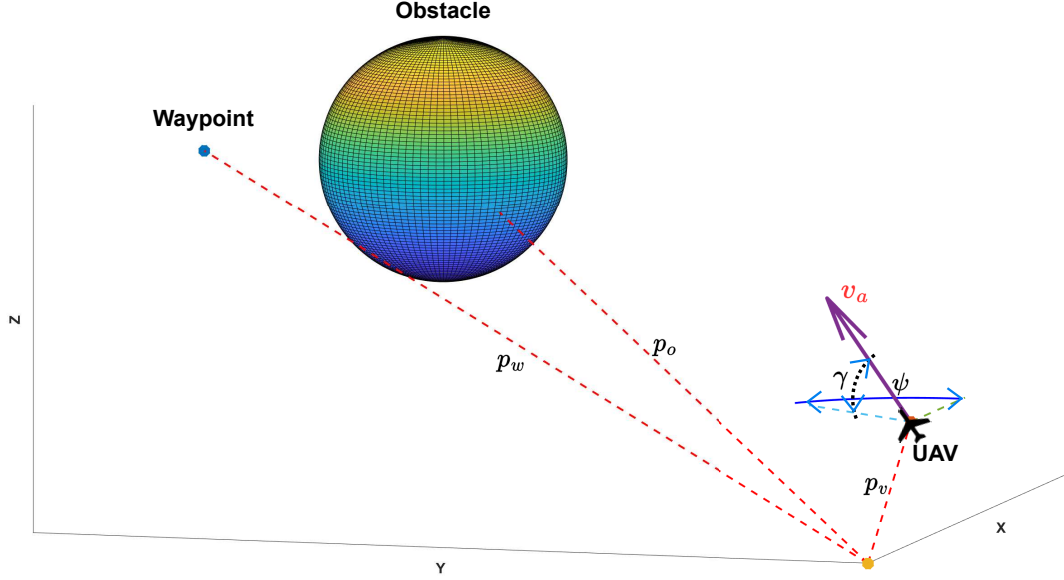


Fig. 1. Illustration of obstacle avoidance scenario

is addressed by modulating the main field without directly adding it with the obstacle field, (ii) the minimum turn rate constraint of nonholonomic vehicles is taken into account by selecting appropriate control parameters, (iii) it is solved in 3D space and (iv) as the generated path is very smooth, the method will be very much helpful for fixed wing UAVs in real-world applications.

II. PROBLEM STATEMENT

A typical architecture of control problems to navigate an autonomous vehicle is a two-layer hierarchical process. The outer control layer generates the linear and the angular speed commands and the inner layer generates corresponding forces and moments. This paper deals with the outer layer of the control to generate speed commands as control inputs of the UAV to reach the desired waypoint while avoiding obstacles on its path.

Let us consider a UAV in a fixed inertial frame of reference in the Cartesian coordinate system as shown in Fig. 1. p_v , p_o and p_w are the position vectors of the vehicle, the obstacle and the desired waypoint, respectively, in the inertial reference frame.

The following kinematic equations govern the UAV.

$$\begin{aligned}\dot{x} &= v_a \cos \gamma \cos \psi \\ \dot{y} &= v_a \cos \gamma \sin \psi \\ \dot{z} &= v_a \sin \gamma \\ \dot{\psi} &= u_\psi \\ \dot{\gamma} &= u_\gamma\end{aligned}\quad (1)$$

where, v_a is the airspeed of the UAV and ψ and γ are the heading and flight path angles, respectively. u_ψ and u_γ

are two control inputs. Now, the objective is to find control variables, u_ψ and u_γ to guide the UAV to reach the desired waypoint while maintaining the minimum safe distance from the obstacle.

III. OBSTACLE AVOIDANCE

Let $p_{ov} = p_o - p_v$ and $p_{wv} = p_w - p_v$ be the positional vectors of the obstacle and the desired waypoint from the UAV, respectively. U_{ov} and U_{wv} are the unit vectors in the direction of p_{ov} and p_{wv} , respectively. In the body-fixed reference frame of the UAV, the x -axis always points towards the desired waypoint. Let U_x be the unit vector along the x -axis. Additionally, we define the z -axis to be perpendicular to the plane formed by U_{ov} and U_{wv} , so,

$$U_z = \frac{U_{wv} \times U_{ov}}{|U_{wv} \times U_{ov}|} \quad (2)$$

where, U_z is the unit vector in the direction of the z -axis of the body fixed reference frame.

And the y -axis is,

$$U_y = \frac{U_z \times U_x}{|U_z \times U_x|} \quad (3)$$

where, U_y is the unit vector in the direction of the y -axis of the body fixed reference frame.

Let θ_x , θ_y , and θ_z represent three ZYX sequence Euler angles to define the current attitude of the UAV with respect to the global reference frame. The rotational matrix, R_{gb} , which transforms coordinates from the body frame to the global frame, is given by:

$$R_{gb} = R_z R_y R_x \quad (4)$$

where,

$$R_z = \begin{bmatrix} \cos \theta_z & -\sin \theta_z & 0 \\ \sin \theta_z & \cos \theta_z & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (5)$$

$$R_y = \begin{bmatrix} \cos \theta_y & 0 & \sin \theta_y \\ 0 & 1 & 0 \\ -\sin \theta_y & 0 & \cos \theta_y \end{bmatrix} \quad (6)$$

$$R_x = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta_x & -\sin \theta_x \\ 0 & \sin \theta_x & \cos \theta_x \end{bmatrix} \quad (7)$$

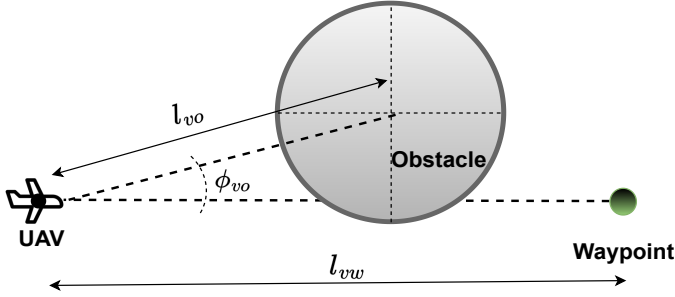


Fig. 2. Illustration of obstacle avoidance scenario

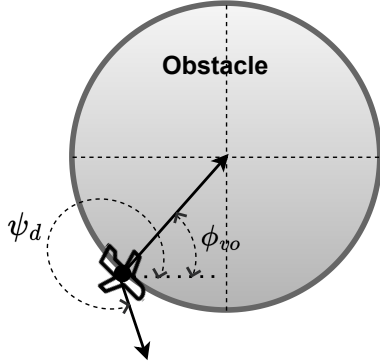


Fig. 3. On the obstacle boundary

The idea behind our approach is to align the XY-plane of the body-fixed reference frame with the global XY-plane. By doing so, a 3D obstacle avoidance problem can be treated as a 2D obstacle avoidance problem. Then, the simplified 2D obstacle avoidance algorithm problem can be addressed to obtain the desired orientation. Finally, this desired vector to be transformed back to the body-fixed reference frame for generating the 3D obstacle free path.

A 2D obstacle avoidance scenario is shown in Fig. 2. Here, ϕ_{vo} represents the LOS angle to the obstacle. r_o is the radius of the obstacle and l_{vo} is the distance between the UAV and the center of the obstacle. l_{vw} is the distance between the UAV and the waypoint as shown in the figure.

The transformed position vector in the global reference frame can be written as,

$$\begin{bmatrix} w_x \\ w_y \\ w_z \end{bmatrix} = R'_x R'_y R'_z * p'_{ov} \quad (8)$$

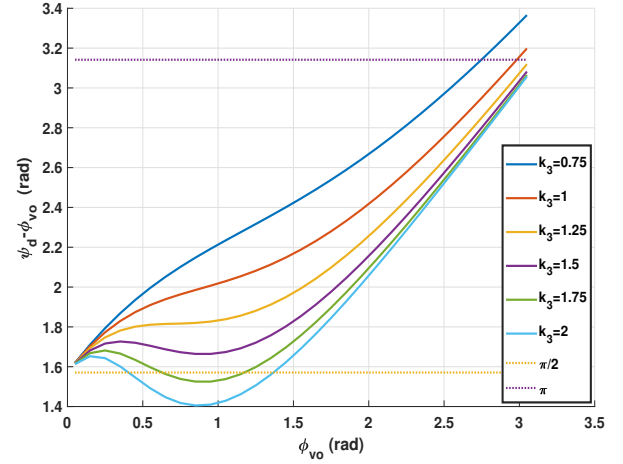


Fig. 4. Angle difference between the desired heading angle and LOS angle to the obstacle, $(\psi_d - \phi_{vo})$ vs. LOS angle to the obstacle, ϕ_{vo} , for different values of k_3 .

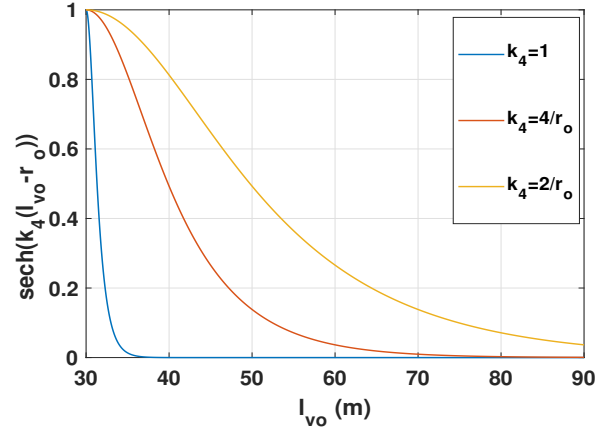


Fig. 5. Profile of $\text{sech}(k_4(l_{vo} - r_o))$ vs. l , for $k_4 = 4/r_o$, $2/r_o$ and $1/r_o$ ($r_o = 30$ m)

From this, ϕ_{vo} is calculated as follows:

$$\phi_{vo} = \arctan(w_y/w_x) \quad (9)$$

The objective is to guide the UAV to the waypoint by avoiding the obstacle in an efficient way. The dynamics of the distance, l_{vo} , is

$$\dot{l}_{vo} = -v_a \cos(\phi_{vo}) \quad (10)$$

A trajectory will be considered as safe for the UAV as long as it moves from its current position towards the waypoint with positive $\dot{l}_{vo}$. If $\dot{l}_{vo}$ is negative, the UAV will be within the collision zone, and it must deviate from its original LOS and follow the angle, ψ_d which is given by the following equation:

$$\psi_d = \frac{\pi}{2} W \text{sech}(k_3 \phi_{vo}) \text{sech}(k_4 (l_{vo} - r_o)) \quad (11)$$

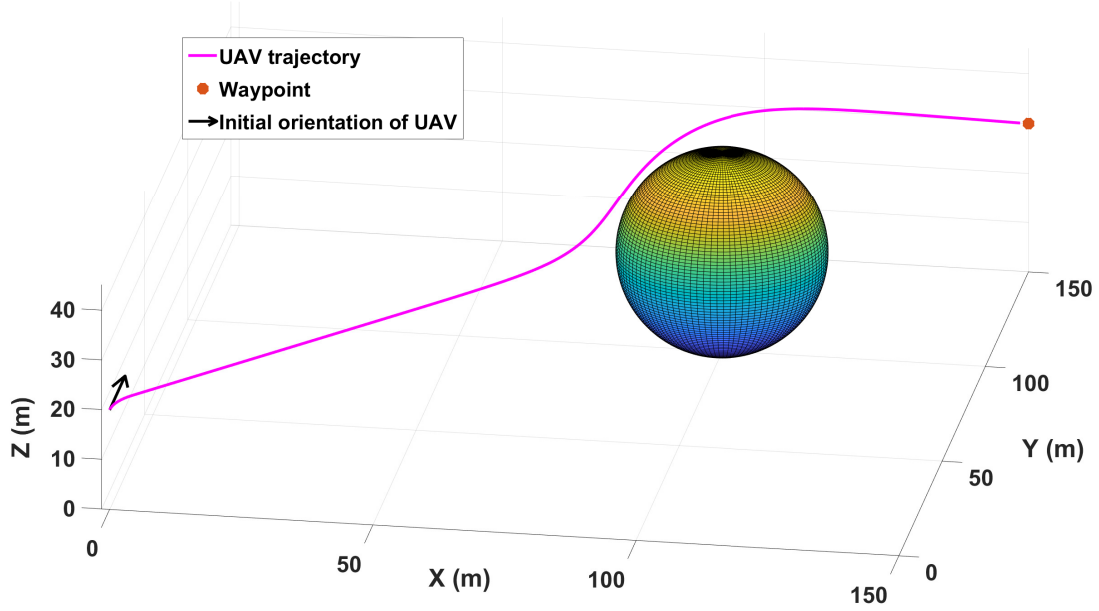


Fig. 6. UAV trajectory for obstacle avoidance

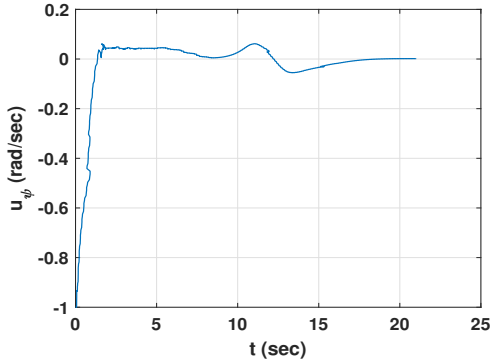


Fig. 7. Profile of control input for single obstacle avoidance case

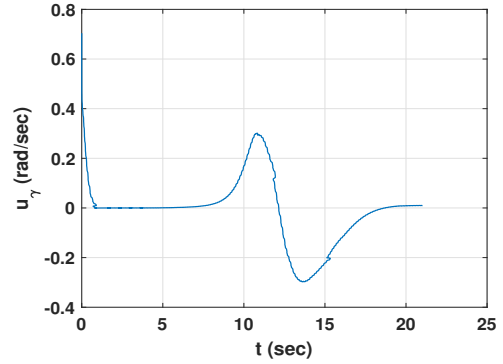


Fig. 8. Profile of control input for single obstacle avoidance case

where,

$$W = \begin{cases} \text{sign}(\phi_{vo}), & \text{if } \phi_{vo} \neq 0 \\ 1, & \text{if } \phi_{vo} = 0 \end{cases}$$

Note that the UAV will only adopt the new LOS angle, ψ_d , when the obstacle is within the detection range of the UAV. The scaling term, $\text{sech}(k_3(\phi_{vo}))$, provides a high value when the obstacle is on the path of the UAV, and a low value when the obstacle is not on the path. Similarly, the scaling term, $\text{sech}(k_4(l_{vo} - r_o))$, offers a high value when the distance between the UAV and the obstacle is small. It is important to select suitable positive values for the gains k_3 and k_4 in order to generate an optimal and safe trajectory.

To determine the value of k_3 , let us consider the scenario where the UAV is located on the boundary of the obstacle, as illustrated in Fig. 3. In this situation, the distance between the UAV and the obstacle is equal to the radius of the obstacle, denoted as $l_{vo} = r_o$. As a result, the term $\text{sech}(k_4(l_{vo} - r_o))$

evaluates to 1. When the UAV is precisely on the boundary of the obstacle, the following condition has to be satisfied for successful obstacle avoidance:

$$\begin{bmatrix} \cos \phi_{vo} & \sin \phi_{vo} \end{bmatrix} \cdot \begin{bmatrix} \cos \psi_d \\ \sin \psi_d \end{bmatrix} \leq 0 \quad (12)$$

$$\cos(\psi_d - \phi_{vo}) \leq 0 \quad (13)$$

It implies,

$$\pi/2 \leq |(\psi_d - \phi_{vo})| \leq \pi \quad (14)$$

or,

$$\pi/2 \leq \left| \frac{\pi}{2} \text{sech}(k_3 \phi_{vo}) - \phi_{vo} \right| \leq \pi \quad (15)$$

Fig. 4 illustrates the relationship between $(\psi_d - \phi_{vo})$ and ϕ_{vo} for various values of k_3 . By examining this plot, it can

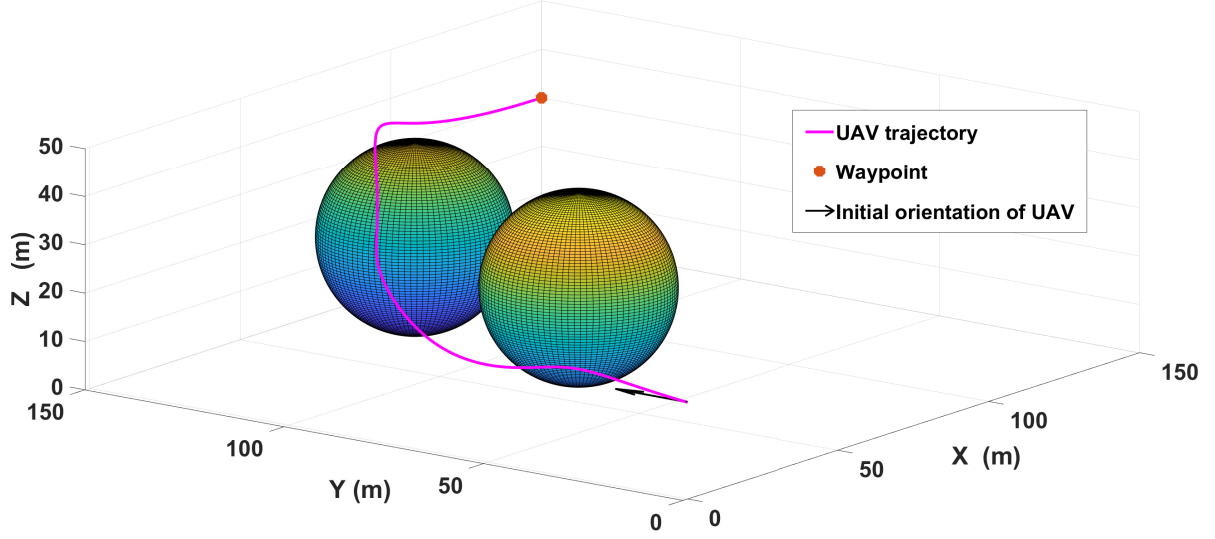


Fig. 9. UAV trajectory for obstacle avoidance

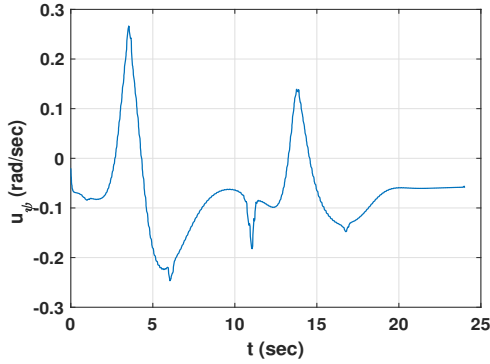


Fig. 10. Profile of control input for multiple obstacle avoidance case

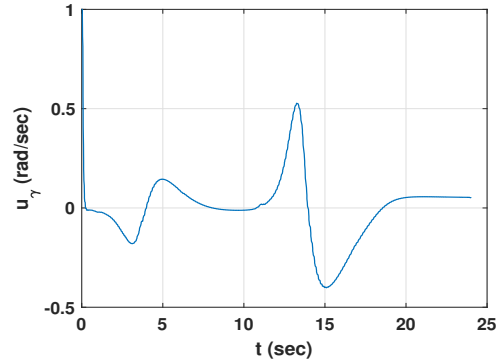


Fig. 11. Profile of control input for multiple obstacle avoidance case

be deduced that $k_3 = 1.5$ fulfills the boundary condition (15), indicating that it is one of the possible safe value for achieving successful obstacle avoidance.

To determine the appropriate value for the gain, k_4 , we consider a scenario where ϕ_{vo} is zero, indicating that the UAV, the obstacle, and the waypoint are aligned in a straight line. In order to avoid the obstacle, the vehicle needs to smoothly deviate from its original path, with the smoothness being controlled by the gain, k_4 . Fig. 5 illustrates the behavior of the function, $\text{sech}(k_4(l_{vo} - r_o))$ as the distance between the UAV and the center of the obstacle (l_{vo}) varies. The graph indicates that when k_4 is chosen as $4/r_o$, around 95% of the change in the final value (which is 1) occurs between r_o and $2r_o$. Similarly, for $k_4 = 2/r_o$, the significant change occurs between r_o and $3r_o$, and for $k_4 = 1/r_o$, it occurs between r_o and $5r_o$. As k_4 decreases, the deviation from the desired trajectory increases, while increasing k_4 reduces the

deviation but may result in an infeasible trajectory. Therefore, selecting an appropriate value for k_4 is necessary for striking a balance between optimality and feasibility.

Using this desired angle we can get the actual desired heading vector in 3D as follows

$$\begin{bmatrix} W_x \\ W_y \\ W_z \end{bmatrix} = R_z R_y R_x \begin{bmatrix} \cos \psi_d \\ \sin \psi_d \\ 0 \end{bmatrix} \quad (16)$$

Using (16), the commanded heading and flight path angles can be obtained as follows

$$\psi_c = \arctan(W_y/W_x) \quad (17)$$

$$\gamma_c = \arctan(W_z/\sqrt{W_x^2 + W_y^2}) \quad (18)$$

Using ψ_c and γ_c , the control inputs can be generated using any convergence guaranteed steering algorithms. Here, the control inputs u_ψ and u_γ are obtained using first order hold loops as follows

$$u_\psi = \alpha_1(\psi_c - \psi) \quad (19)$$

$$u_\gamma = \alpha_2(\gamma_c - \gamma) \quad (20)$$

In scenarios involving multiple obstacles, the desired heading angle can be determined using the following equation:

$$\psi_d = \sum_{i=1}^m \frac{\pi}{2} W_i \text{sech}(k_3 \phi_{voi}) \text{sech}(k_4 (l_{voi} - r_{oi})) \quad (21)$$

Here, m represents the total number of obstacles, while ϕ_{voi} and l_{voi} refer to the line-of-sight angle and the distance between the UAV and the i^{th} obstacle, respectively. It is important to note that if two obstacles overlap, they should be treated as a single obstacle.

IV. SIMULATION RESULTS

This section presents simulation results for waypoint following in an obstacle filled environment to validate the proposed vector field guidance algorithm. Following data are considered for the simulation. The desired waypoints is (150, -150, 30) m, the static obstacle is located at (75, 125, 20) and for multiple obstacle avoidance case another obstacle is located at (30, 50, 30) with radius of 20 m. The UAV is moving with a constant linear speed of 10 m/sec to reach the waypoint from its initial position, (0, 0, 20) m, with initial heading angle $\pi/2$ and flight path angle of 0 rad. The maximum and the minimum angular speeds of the UAV are 1 rad/sec and -1 rad/sec, respectively. Fig. 6 shows the trajectory of the UAV for the single obstacle avoidance. It can be seen from the figure that the path is very smooth and avoids the obstacle to reach the goal. Control inputs (u_ψ and u_γ) of the UAV corresponding to this case are depicted in Fig. 7 and 8, respectively. Fig. 9 shows the trajectory of the UAV for multiple obstacle avoidance case. And corresponding control inputs of the UAV are depicted in Fig. 10 and 11, respectively.

V. CONCLUSION

A novel method for guiding UAVs in a 3D obstacle filled environment has been presented in this work. The singularity problem of traditional artificial potential field based algorithms has been eliminated by modulating the main field separately from the obstacle field. Furthermore, the nonholonomic constraints of the vehicle during obstacle avoidance has been included by selecting appropriate control parameter. After generating vector fields, the streamlines derived from them have served as feasible trajectories of the UAV. The numerical simulations have been presented to demonstrate the efficacy of the presented approach in achieving obstacle avoidance and successful UAV navigation. To enhance the practicality and versatility of the proposed

approach different geometric shapes of obstacles will be considered in future.

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